

AWS Cloud Internship Project

FINAL PROJECT REPORT

Custom VPC + EC2 Web Application + S3 Storage + IAM Roles + CloudWatch Monitoring + Automated Backup and Cost Control

AWS Cloud Internship Submission

Project at a Glance

Item	Details
AWS Region	Asia Pacific (Mumbai) — ap-south-1
VPC	InternshipProject-VPC — 10.0.0.0/16
Subnet	InternshipProject-PublicSubnet — 10.0.1.0/24
EC2	Ubuntu Linux t3.micro, 8 GiB gp3
Application	Apache2 Student Portal + /health endpoint
S3	Private project bucket with IAM role access
Monitoring	CloudWatch CPU and status-check alarms
Backup	EBS snapshot and restore validation
Cost	AWS Budget + tagging + cleanup

Sensitive identifiers have been redacted in the embedded evidence screenshots.

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1. Project Overview

This project demonstrates a secure, cost-conscious AWS environment for an internship/student web application. The implementation combines a custom VPC, public subnet, Internet Gateway, Linux EC2 web server, S3 storage, IAM roles and policies, CloudWatch monitoring, SNS notifications, EBS backup/restore, AWS Budget controls, and web-access-log analysis.

2. Project Objectives

The project objectives were to build a custom VPC and public subnet, deploy a Linux web server, apply least-privilege IAM controls, integrate EC2 with S3 without hardcoded credentials, monitor the EC2 instance with CloudWatch, send alerts through SNS, create an EBS snapshot and validate restoration, control cost with AWS Budgets, and analyze a sample web-access-log dataset.

3. AWS Architecture

Traffic enters through the Internet Gateway into the custom VPC and public subnet, where the EC2 web server hosts the Student Portal. The EC2 instance uses an IAM role for controlled S3 access. CloudWatch monitors the instance and sends alarm notifications through SNS. EBS provides the root volume and a snapshot is retained as the backup point. AWS Budget and resource tagging support cost control.

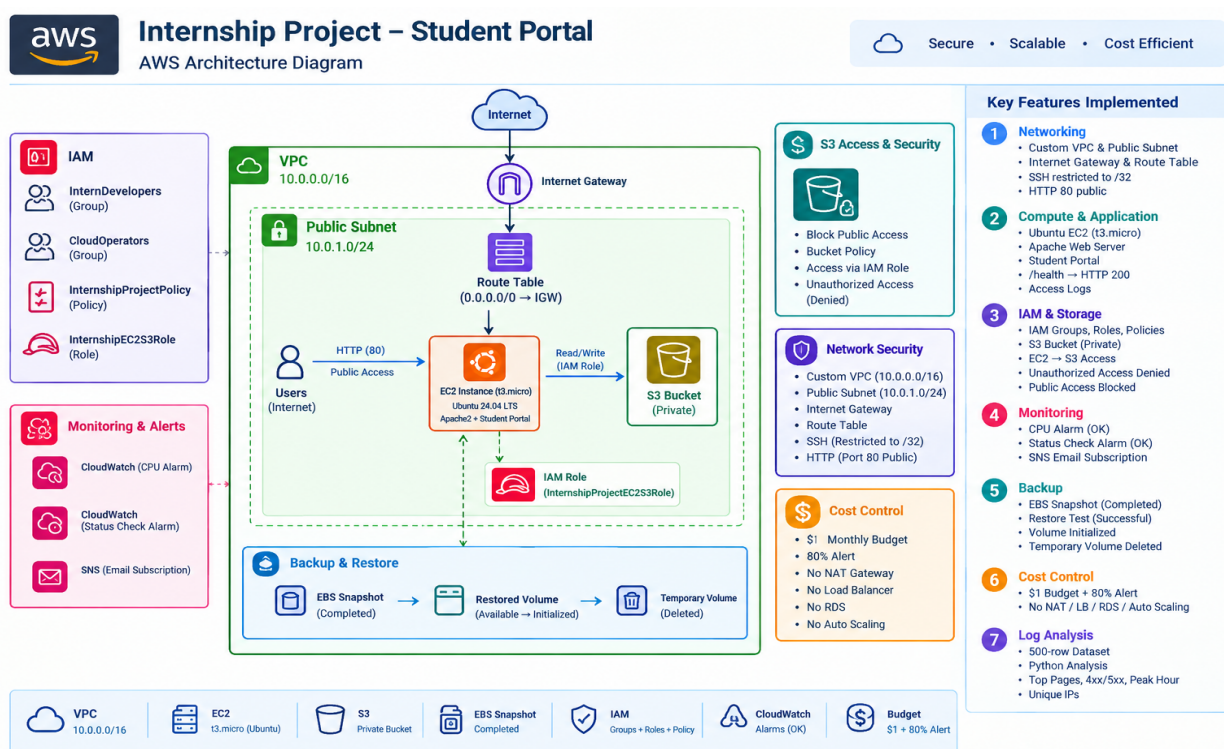


Figure 1 — AWS project architecture overview

4. IAM & Security

Root MFA was enabled and no active root access keys were retained. The project uses the InternDevelopers group and a customer-managed InternshipProjectPolicy. The EC2 instance uses InternshipEC2S3Role with an inline policy that allows access only to the designated project S3 bucket. The project also validated that access to an unrelated bucket is denied.

5. VPC & Networking

The custom network uses VPC CIDR 10.0.0.0/16 and a public subnet 10.0.1.0/24 in ap-south-1a. An Internet Gateway is attached and the public route table sends 0.0.0.0/0 to the Internet Gateway. The web security group allows HTTP on port 80 and restricts SSH on port 22 to the user's IP address.

6. EC2 & Student Portal

The project web server is an Ubuntu Linux t3.micro instance with an 8 GiB gp3 root volume, the project key pair, the custom VPC/subnet, the hardened security group, and the InternshipEC2S3Role instance profile. Apache hosts the Student Portal under /student-portal/. A /health endpoint returns HTTP 200 to provide a simple operational health check.

7. S3 Storage & IAM Integration

The project bucket is private with Block Public Access enabled and bucket-owner-enforced ACL behavior. The EC2 role can list the designated bucket and read/write objects within it. A test object named s3-test.txt was uploaded from the EC2 instance using the AWS CLI without embedding long-term access keys on the server.

8. CloudWatch Monitoring & SNS

Two CloudWatch alarms were configured: InternshipProject-EC2-HighCPU for CPU utilization above 70 percent and InternshipProject-EC2-StatusCheck for failed status checks. Both alarms were connected to the InternshipProject-Alerts SNS topic, and the email subscription was confirmed. The alarms were shown in OK state during verification.

9. EBS Backup & Restore

A snapshot named InternshipProject-EC2-Backup was created from the EC2 root EBS volume and reached Completed state. A temporary 8 GiB gp3 volume was restored from the snapshot in the same Availability Zone and initialized successfully. The temporary restored volume was then deleted while the snapshot was retained.

10. Web Log Analysis

A 500-row web-access-log.csv dataset was analyzed using Python. The analysis calculated total requests, top pages, 4xx and 5xx error counts, unique IP addresses, and the peak traffic hour. Results were: 500 total requests, 21 4xx errors, 6 5xx errors, 35 unique IP addresses, and peak traffic from 16:00–16:59 with 63 requests.

11. Cost Control & Free Tier

The implementation intentionally avoided high-cost services such as NAT Gateway, load balancers, RDS, and Auto Scaling. A recurring \$1 monthly AWS Budget was configured with an alert at 80 percent actual spend. Resources were tagged with Project=AWS-Internship, and the temporary restored EBS volume was removed after validation.

12. Testing & Validation

Validation covered the live Student Portal, the /health HTTP 200 endpoint, EC2-to-S3 role access, denial of unrelated S3 access, CloudWatch alarm state, SNS subscription confirmation, EBS snapshot completion and restoration, and an EC2/resource check. These tests provide evidence that the core security, networking, application, storage, monitoring, backup, and cost-control components were functioning as intended.

13. Challenges & Solutions

Key implementation challenges included avoiding accidental changes to the separate Linux-practice EC2 instance, keeping the project resources inside the intended custom VPC, configuring least-privilege IAM rather than broad permissions, and verifying backup restoration without leaving unnecessary resources running. The final implementation addressed these through explicit resource naming, scoped policies, validation commands, and cleanup checks.

14. Final Outcome & Skills Demonstrated

The project demonstrates practical skills in AWS IAM, VPC networking, EC2 administration, Apache web hosting, S3 integration, IAM role-based authentication, CloudWatch monitoring, SNS notifications, EBS snapshots, AWS Budgets, Linux administration, AWS CLI usage, Python log analysis, troubleshooting, security validation, and cost-aware cloud design.

15. Future Improvements & Conclusion

Future improvements could include a private application tier, automated deployment through CI/CD, Infrastructure as Code with Terraform, centralized log delivery, stronger application authentication, and a multi-AZ design when cost constraints are relaxed. For the internship scope, the completed project provides a compact end-to-end cloud environment with security, monitoring, backup, testing, and cost-control evidence.

Evidence Screenshots

Sanitized evidence plates are included below. Redactions preserve the technical evidence while hiding sensitive identifiers.

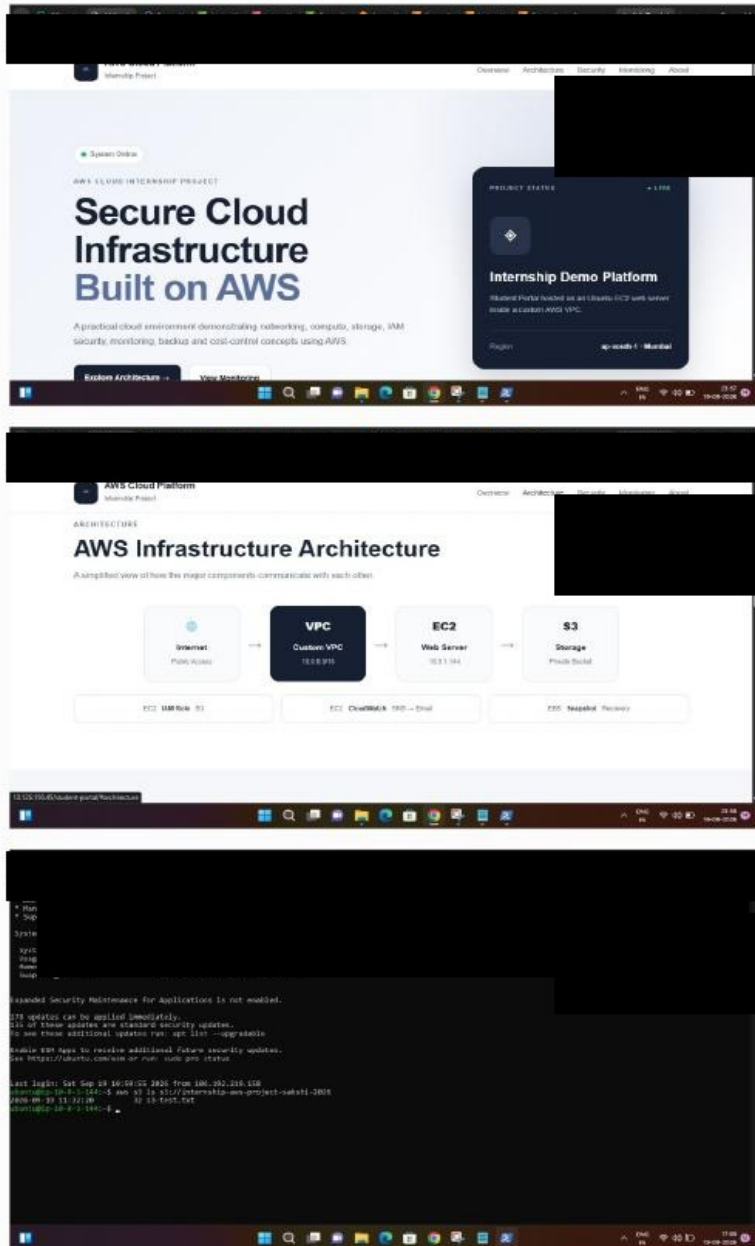


Figure 2 — Evidence Plate 1 — Student Portal landing page, AWS architecture page, and EC2/application terminal evidence.

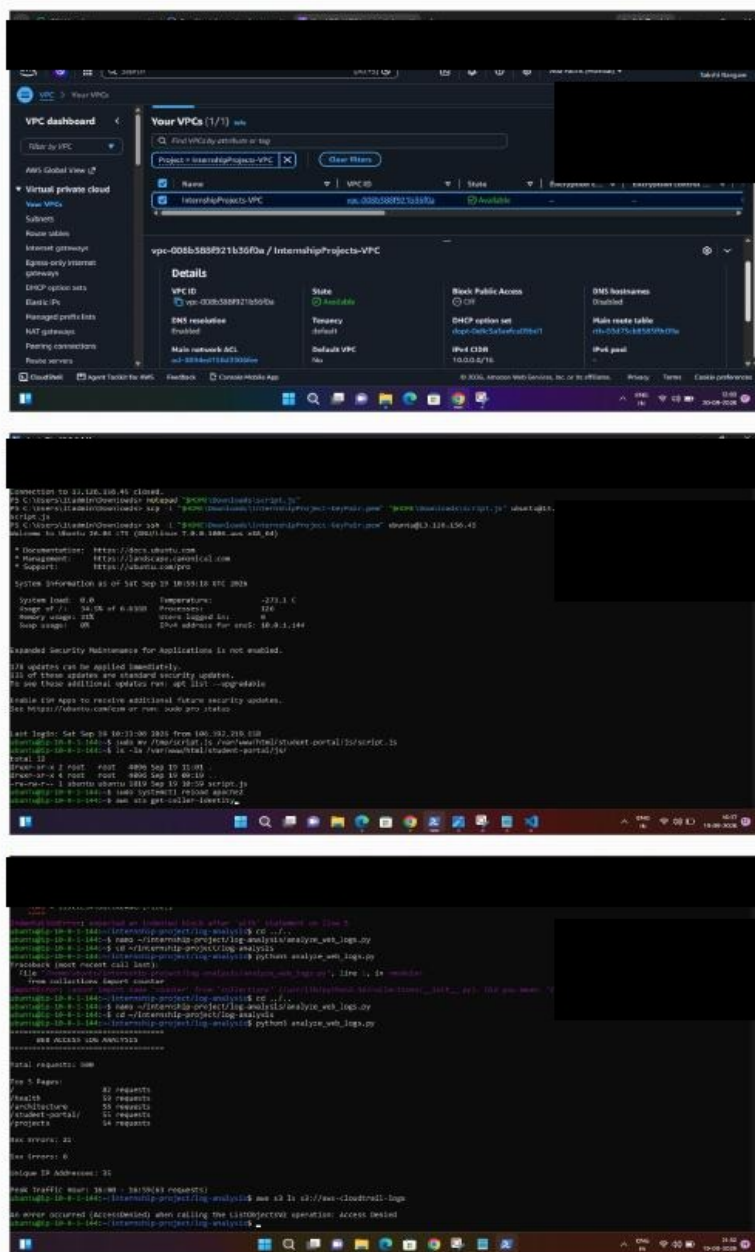


Figure 3 — Evidence Plate 2 — Custom VPC configuration, Linux/Apache validation, and Python web-log analysis output.



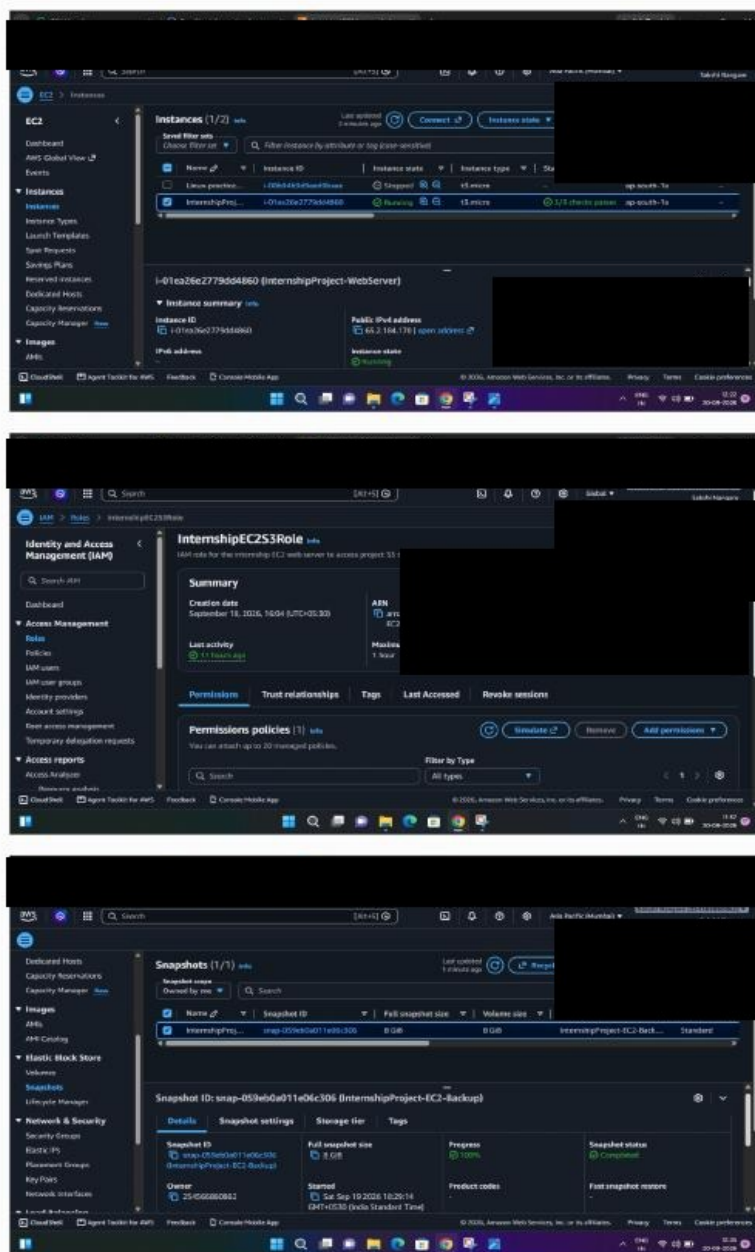


Figure 5 — Evidence Plate 4 — EC2 instance and related compute configuration evidence.

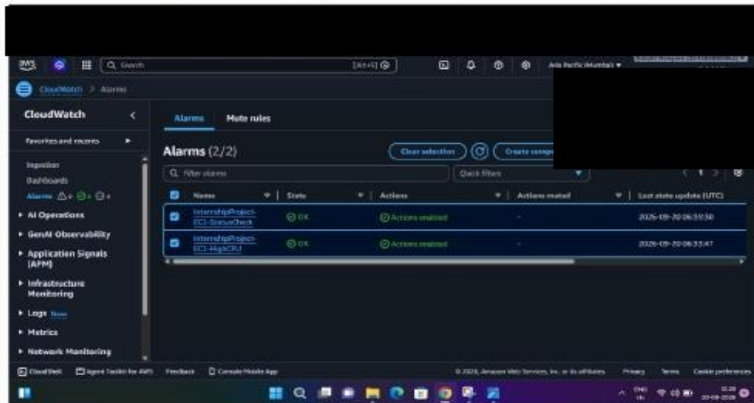
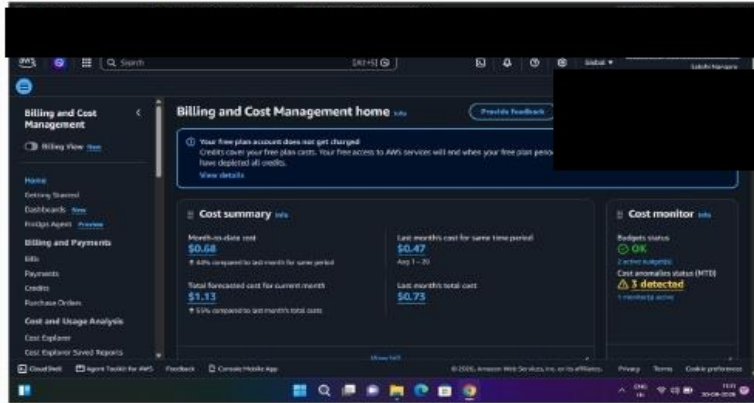
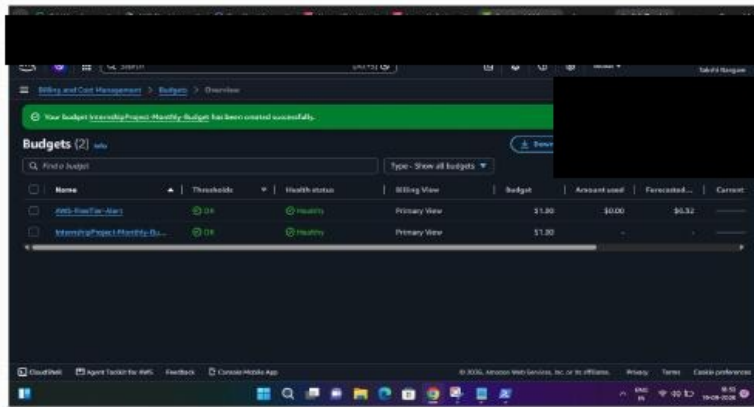


Figure 6 — Evidence Plate 5 — AWS Budget and cost-control evidence.

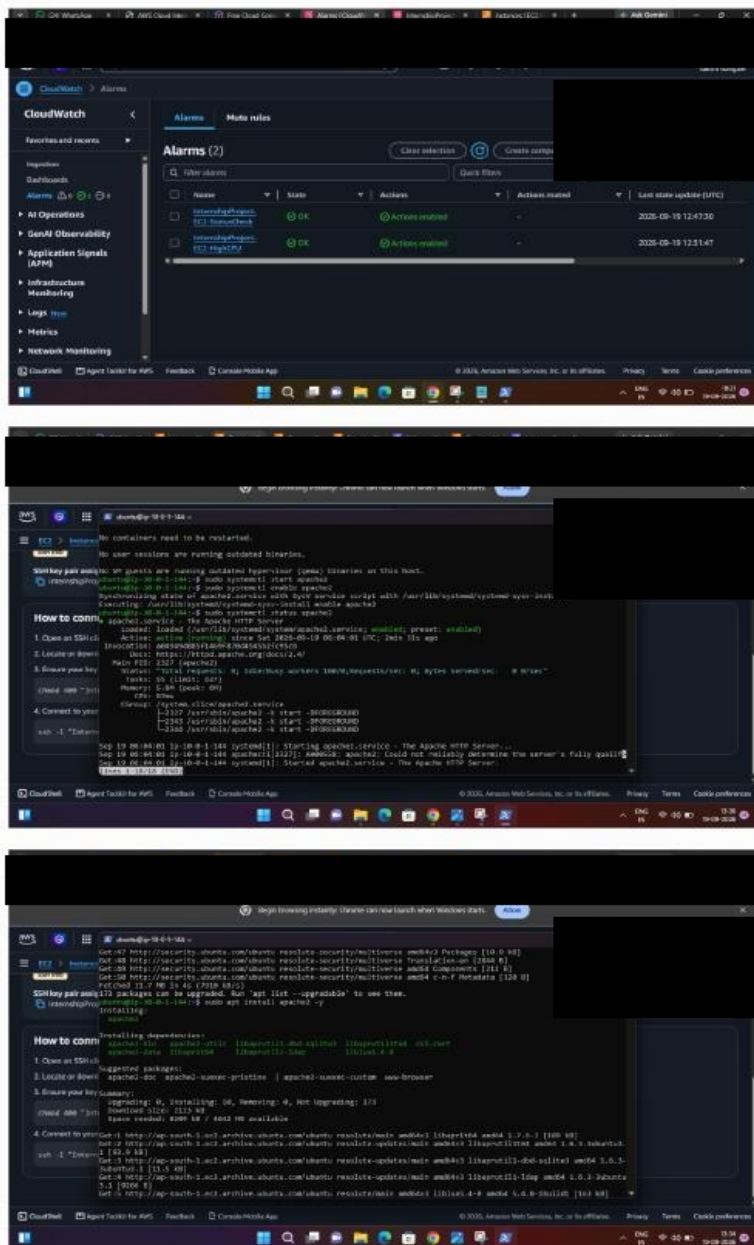


Figure 7 — Evidence Plate 6 — CloudWatch alarm configuration and monitoring evidence.

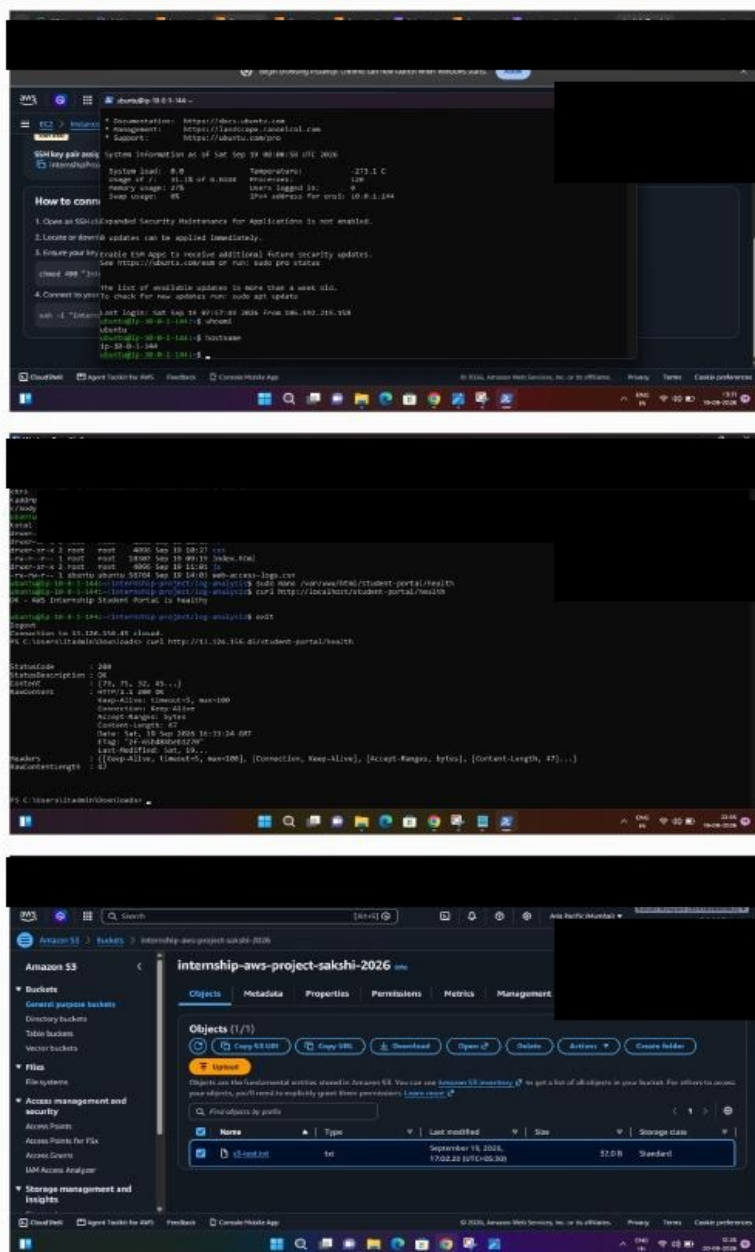


Figure 8 — Evidence Plate 7 — EC2 connection/application and S3-related terminal evidence.

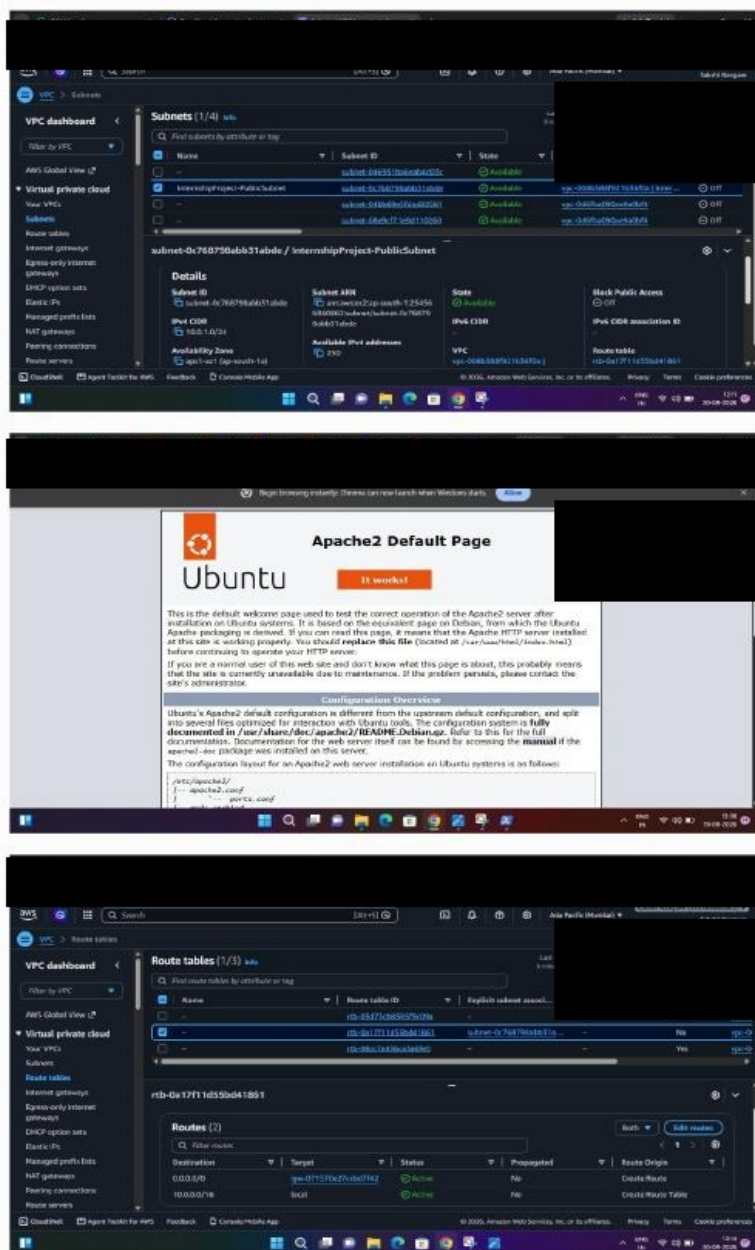


Figure 10 — Evidence Plate 9 — Public subnet and VPC networking configuration evidence.